

Jay Nathan

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Objective: Seeking an entry level position as a hardware design engineer for Computer Architecture and VLSI systems for HPC & AI/ML acceleration. My research interests lie at the intersection of RTL design, CAD flows, and AI/ML hardware co-design. I have chip design, fabrication and systems software development experience.

Education:

University of Southern California, Los Angeles

Expected May 2027

(Master of Science in Electrical & Computer Engineering)

University of Illinois, Urbana-Champaign

December 2025

(Bachelor of Science in Electrical Engineering w/ Minor in Computer Science)

Skills Summary:

- **Hardware:** Computer Organization & Design, Digital Systems, Digital IC Design, VLSI System Design, IC Device Theory and Fabrication, Semiconductor Devices, Digital Signal Processing, Analog Signal Processing, Electronic Circuits
 - **Software:** Operating Systems, Data Structures & Algorithms, Computer Systems & Programming
 - **AI/ML:** Artificial Intelligence, Sensor Fusion, Applied Machine Learning, Deep Learning for Computer Vision Systems
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Work Experience:

Graduate Researcher: Ray Tracing Wireless Channel Simulator

[Wireless Devices & Systems \(WiDeS\) Group](#), USC (Advisor: Prof. [Andreas F. Molisch](#) | Supervising Ph.D. Student: [Bassel Modad](#)) Jan 2026 - Ongoing

- Currently developing a ray-tracing engine in C++ for simulating 3D wireless propagation, analyzing and enhancing implementations of specular reflection, UTD edge diffraction, and diffuse scattering across complex triangle-mesh geometries in 6G networks.
- Collaborating on ML-based channel modeling efforts by validating deterministic ray-tracing outputs against neural network approximations for improved simulation speed-accuracy tradeoffs.
- Developing GPU-accelerated ray-tracing infrastructure to generate high-fidelity RF propagation datasets used for training deep learning path-loss prediction models.
- Supporting next-generation wireless system research by enabling high-fidelity channel modeling infrastructure used for ML-based path-loss modeling and RF performance evaluation.

Graduate Researcher: LLM-Based Verilog Design & Debugging Agent (CVDP)

[Energy Efficient Secure Sustainable Computing Group](#), USC (Advisor: [Prof. Peter Beerel](#)) Jan 2026 - Ongoing

- Developing an LLM based agent for automated **Verilog RTL generation and debugging**, targeting hardware design and verification workflows.
- Redesigning the agent's **planning module** to enable reliable multi-step reasoning over code edits, simulation results, and testbench failures.
- Integrating **Verilog AST based program analysis** to localize RTL errors and guide structured, syntax-aware fixes rather than text-only edits.
- Designing a **persistent memory system** for the agent to retain past failures, fixes, and error patterns across debugging iterations.

Undergraduate Researcher: [Center for Auto Construction & Manufacturing at Scale](#) (prev [AUVSL](#)) May 2023-2025

Worked with the team to design and implement outdoor robot localization solution leveraging Ultrawide-Band Antennas by **writing Python & C++ packages**, deployed on the Clearpath Husky & Jackal platforms.

- Collaborated with cross-disciplinary teams of 5–7 researchers integrating ML and embedded hardware pipelines; contributed to both algorithmic and systems-level optimization.
- **Developed PyTorch scripts** to tune Adaptive Neural Fuzzy Inference System parameters, worked on a **custom Unscented Kalman Filter (UKF)** for sensor fusion with UWB, IMU, and wheel encoders, to reduce drift
- Utilized Python to write localization software and scripts **utilizing popular libraries to analyze and process sensor data**.

HARDWARE SYSTEMS PROJECT EXPERIENCE

Advanced VLSI System Design Project ([ECE 427](#)):

May 2025 – Dec 2025

(Sponsored by [APPLE](#), in collaboration with [TSMC](#), [MUSE](#) semiconductor, and UIUC Grainger School of Eng.)

Collaborated with a team of six to build a mixed-signal chip that integrates a True Random Number Generator (TRNG) with hardware-accelerated AES encryption. Responsible for RTL implementation, physical design, and chip-level integration. The project followed a full RTL-to-GDSII flow within 1 mm²/50 mW constraints, including synthesis, place-and-route, and verification using Synopsys Design Compiler, Cadence Innovus, and Calibre. This design was sponsored and selected by Apple and TSMC for fabrication, validating technical rigor and real-world relevance for secure, low-power IoT applications.

Senior Research Project & Senior Thesis ([ECE 496](#) + [ECE 499](#)):

Jan 2025 – Nov 2025

Currently designing a hardware accelerator for a neuromorphic processor under Professor [Anu Aggarwal](#) at UIUC's Grainger College of Engineering. Focused on low-power architectures to accelerate AI/ML workloads such as gradient descent and matrix multiplication, with responsibilities spanning design, architectural exploration, and optimization for parallelism and energy efficiency.

Applied Machine Learning Systems ([ECE 445](#))

Dec 2025

In this System Design Project under the guidance of Professor Cunjiang Yu, my team is implementing a low-cost, modular diagnostic device for rural healthcare clinics. The system integrates analog and digital inputs into a microcontroller-based acquisition platform, with embedded firmware managing real-time sensor processing and AI/ML techniques interpreting the data. By unifying circuit design, microcontroller control and ML interface, the project demonstrates my ability to translate pressing market needs into end-to-end system solutions spanning electrical engineering, computer engineering, and computer science.

RV32IM N-Way Superscalar Out-of-Order Explicit Register Renaming-Based CPU ([ECE 411](#)) May 2024

- Designed & verified from scratch an out-of-order N-way parameterizable processor for RISC-V 32-bit ISA with M-extension, achieving IPC of 0.51 on standard benchmark taking 21.7 mW of power at clock frequency 325 MHz.
- Implemented speculative branching with branch prediction overriding, & perceptron and Gshare branch predictors.
- Implemented cache features including next-line & stride prefetchers, post-commit store buffer with write coalescing.
- Integrated Synopsys IPs including sequential divider and wrote Dadda advanced multiplier and shift-add multiplier.
- Wrote full processor in SystemVerilog, simulated with Synopsys VCS, debugged with Synopsys DC, Verdi, Spike, RVFI, Lint, Functional Coverage, Used Python to generate test programs, built toolchains with bash, GNU make.

Bitsliced RV32I CPU in Cadence Virtuoso ([ECE 425](#) and [ECE 482](#))

Apr 2025

- Authored FreePDK 45nm standard library (logic, muxes, flip-flops) with schematic, layout, and Liberty timing views.
- Manually laid out in Virtuoso a complete 32-bit RISC-V RV32I pipeline using a bitsliced methodology in a hierarchical library, complete with ALU, write-low/read-high register file, barrel-shifter, and cascaded comparator.
- Verified extracted netlist with SystemVerilog harness, passing reference, and self-written RV32I assembly suites.
- Wrote Tcl scripts for Innovus to make 2nd processor in addition to the manual layout to automate the layout process & minimize area for the entire processor, including floorplanning, placement, routing, & DRC/LVS checks.

IC Fabrication – ([ECE 340](#) and [ECE 444](#)) Course-Based Project at UIUC

Dec 2024

- Independently processed a silicon wafer from start to finish in an industrial cleanroom environment, performing every step—from RCA cleaning and oxidation to photolithography, etching, doping, and final metal lift-off.
- Utilized a five-mask sequence to fabricate complex devices (MOSFETs, BJTs, diodes, logic gates).

AI/ML PROJECT EXPERIENCE

Machine Learning & AI Research – [CACMS](#) / [AUVSL](#)

Jan 2024 – May 2025

- Researched and adapted the **MOSTS trajectory-smoothing algorithm** for multi-robot navigation, improving stability and reducing path error.
- Built automated pipelines to generate and preprocess **4,000+ synthetic terrain images** (snow, mud, rocky) with diffusion models, rotation/cropping, and **HDF5 dataset packaging with labels/timestamps**.
- Developed and trained **CNN models (EfficientNet-B0, MobileNetV3)**, achieving **99.3% accuracy** and **95.8% recall on snow terrain**, with performance on external datasets.
- Implemented an **Unscented Kalman Filter** for real-time sensor fusion (UWB + IMU + encoder), and optimized model deployment by exporting to **ONNX Runtime with DirectML**, reducing inference latency from **16.3 ms** → **7.1 ms (~2.3× faster)** for **140 FPS real-time performance**.
- Developed and deployed quantized LLM-based agents and Graph RAG with a knowledge graph for autonomous networking on NVIDIA A30/A100 GPU/DPU Bluefield platform. Inference deployed on Ubuntu-based edge nodes accelerated by NVIDIA GPUs, DPUs, and ARM/Intel CPUs, cutting remediation time by 45% and reducing MTTR by 30%.
- My early research trajectory was shaped under the guidance of the late **Professor William [Robert “Bob” Norris](#)** at these laboratories, where I first engaged in AI/ML research for autonomous systems. His mentorship instilled the research curiosity that continues to guide my work.

Artificial Intelligence Algorithms & Applications ([CS 440](#))

Apr 2024

- Implemented **Naive Bayes classifier** for text classification with smoothing and feature selection.
- Developed *search algorithms* (*DFS, BFS, A, UCS*)* for pathfinding and planning.
- Built **HMM-based part-of-speech tagger** using Viterbi decoding, improving sequence labeling accuracy.
- Designed a hardware-accelerated Deep Neural Network for AMD PYNQ-Z2 FPGA using Xilinx Vivado HLS.
- Designed **Q-learning reinforcement learning agent** in Python/PyTorch for Atari-style environments.

Applied Machine Learning Systems ([CS 441](#))

Apr 2024

- Applied ML techniques to **classification, regression, clustering, and graphical models** across vision and language domains.
- Implemented **logistic regression and SVM classifiers** with regularization and cross-validation.
- Designed and trained **CNNs in PyTorch/TensorFlow**, achieving ~79% accuracy on CIFAR-10-like datasets.
- Applied **gradient descent optimization** (SGD, momentum, learning rate scheduling) for faster convergence.
- Explored **advanced ML techniques**, including **Mean Field methods, Variational Auto-Encoders (VAEs), and Hidden Markov Models (HMMs)**.

SOFTWARE PROJECT EXPERIENCE

Design and development of Unix-Like Operating System

Dec 2023

- Engineered a UNIX-like, single-core kernel for 32-bit hardware fully from scratch using C and x86 Assembly.
 - Implemented a paging-only virtual memory system, writable file system, software context switching, terminal switching, and hardware interrupts and exception handling.
 - Developed device drivers for keyboard, mouse, real-time clock, & built interactive shell for executing system calls.
 - Used advanced GDB techniques to efficiently debug and resolve system-level issues.
 - Emulated and tested the OS in a QEMU virtual environment, ensuring safe and efficient debugging.
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Technical Skills:

Languages: C++, C, C#, Python, Java, Matlab, x86 Assembly, RISC-V ISIP, System Verilog

Hardware: (Board Bringup/Programming):

AMD/Xilinx FPGA (VIVADO environments), Intel/Altera Stratix FPGA/SoC series (RISC-V CPU designs)

Tools/Environments: CUDA, DOCA, GDB, Linux, Bash/Shell, TCL, Git, Docker, Jenkins, Make, Vim, QEMU, Anaconda/Jupyter, Docker, Github/Gitlab, LaTeX

Libraries: Tensorflow, Keras, PyTorch, NumPy, SciPy, Scikit-Learn, OpenCV, Pandas

EDA/RTL DV: SystemVerilog, SV Coverage, Intel Quartus, AMD Xilinx, Chisel, HLS, PyRTL, Veryl, Synopsys DC & VCS, Verdi, Spyglass, ModelSim/QuestaSim, Verilator, Chipyard, Cadence Virtuoso & Innovus, Cocotb, Gem5, UVM

Interfaces & Test Methodologies: AXI, UART, SPI, I2C, I2S, CAN, JTAG, ATPG, BIST, ATE, TCP/IP

Other: ROS 1/2, RViz, Gazebo, RPi, Arduino, Simulink, Unity, Eagle CAD, Autodesk Inventor, SolidWorks

- Hardware-Software Co-Design (ML accelerators • RISC-V ISA • Neuromorphic systems)
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References:

- [Professor Anu Aggarwal](#) (ECE Department, Formerly Grainger College of Engineering, UIUC/Now at CalPoly)
- [Paul J Froeter](#) (ECE Department/Nanotechnology Lab, The Grainger College of Engineering, UIUC)
- [Rajiv Sharma](#) (Industry mentor) LPU (Language Processing Unit ASIC) Senior Engineering Lead at [GROQ](#)
- [I-Chen Sang](#), Post Doc Research Assistant, [CACMS](#), The Grainger College of Engineering UIUC)
- [Hsiao-Yu Chen](#), Ph.D candidate [CACMS](#), The Grainger College of Engineering UIUC)